

NCDF Report for Kisii, Nyamira and Kajiado

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Introduction

The National Government Constituencies Development Fund (NG-CDF) is a key mechanism for financing grassroots development projects in Kenya. This study focuses on allocations in Kisii, Nyamira, and Kajiado counties between 2014 and 2026. The aim is to analyze allocation trends, compare constituency performance, and assess growth rates.

A Shiny dashboard was developed to provide stakeholders with interactive visualizations of allocations. For reporting purposes, this Quarto document integrates static plots and tables that summarize the findings. While allocations are clearly tracked, the analysis also highlights the gap in linking financial flows to actual projects implemented, which is essential for accountability and impact assessment.

Methodology

The analysis was conducted using R and several supporting packages:

tidyverse for data manipulation and wrangling.

scales for formatting numeric outputs.

ggplot2 for data visualization.

knitr for static tables.

```
# Load required packages
library(tidyverse) # includes dplyr, ggplot2, tidyr, readr, etc.
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```

v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```

```
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

```
discard
```

The following object is masked from 'package:readr':

```
col_factor
```

```
library(knitr)
```

Steps Step 1: Load Data Load the dataset containing NCDF allocations.

```

NCDF <- read.csv("NEW_NGCDf_DATA_28-JAN_2026.csv", sep = ',', header = TRUE)
NCDF$County <- trimws(NCDF$County)

```

Step 2: Filter Counties of Interest Restricted the dataset to Kisii, Nyamira, and Kajiado.

```

NCDF_County <- dplyr::filter(
  NCDF,
  County %in% c("Kisii", "Kajiado", "Nyamira"),
  County != 0
)
head(NCDF_County);

```

	County	Constituency	Sub.County	X	IEBC_CN	IEBC.CC	IEBC.CN
1	Kajiado	Kajiado North	Kajiado North	NA	KAJIADO NORTH	34	KAJIADO
2	Kajiado	Kajiado Central	Kajiado Central	NA	KAJIADO CENTRAL	34	KAJIADO
3	Kajiado	Kajiado East	Kajiado East	NA	KAJIADO EAST	34	KAJIADO
4	Kajiado	Kajiado West	Kajiado West	NA	KAJIADO WEST	34	KAJIADO
5	Kajiado	Kajiado South	Kajiado South	NA	KAJIADO SOUTH	34	KAJIADO
6	Kisii	Bonchari	Bonchari	NA	BONCHARI	45	KISII

	IEBC.CONSC	VOTERS	ColCODE	ConstNAME	X2014	X2015	X2016
1	183	134880	183	Kajiado-North	69115807	98758692	104582462
2	184	65823	184	Kajiado-Central	74542261	106979912	113362984
3	185	116336	185	Kajiado-East	70219170	100430316	106367808
4	186	75173	186	Kajiado-West	77183331	110981202	117636488
5	187	71061	187	Kajiado-South	78964640	113679933	120518817
6	261	64630	261	Bonchari	74868981	107474902	113891649

	X2017	X2018	X2019	X2020	X2021	X2022	X2023	X2024
1	81896552	98189655	109040876	137367724	137088879	137088879	145087603	166593720
2	81896552	98189655	109040876	137367724	137088879	137088879	145087603	175361810
3	81896552	98189655	109040876	137367724	137088879	137088879	145087603	175361810
4	81896552	98189655	109040876	137367724	137088879	137088879	145087603	175361810
5	81896552	98189655	109040876	137367724	137088879	137088879	145087603	175361810
6	81896552	98189655	109040876	137367724	137088879	137088879	138215033	175361810

	X2025	X2026
1	179898372	192613248
2	179441955	192613248
3	179640955	192613248
4	179441955	192613248
5	179441955	192613248
6	170469857	182982585

Step 3: Reshape Data Pivot yearly allocations into long format for easier analysis.

```
NCDF_long <- NCDF_County %>%
  pivot_longer(cols = starts_with("X"),
               names_to = "Year",
               values_to = "Allocation") %>%
  mutate(Year = as.numeric(sub("X", "", Year))) %>%
  filter(!is.na(Year))
head(NCDF_long);
```

A tibble: 6 x 12

	County	Constituency	Sub.County	IEBC_CN	IEBC.CC	IEBC.CN	IEBC.CONSC	VOTERS
	<chr>	<chr>	<chr>	<chr>	<int>	<chr>	<int>	<int>
1	Kajiado	Kajiado North	Kajiado North	KAJIADO~	34	"KAJIA~	183	134880
2	Kajiado	Kajiado North	Kajiado North	KAJIADO~	34	"KAJIA~	183	134880

```

3 Kajiado Kajiado North Kajiado North KAJIADO~      34 "KAJIA~      183 134880
4 Kajiado Kajiado North Kajiado North KAJIADO~      34 "KAJIA~      183 134880
5 Kajiado Kajiado North Kajiado North KAJIADO~      34 "KAJIA~      183 134880
6 Kajiado Kajiado North Kajiado North KAJIADO~      34 "KAJIA~      183 134880
# i 4 more variables: ColCODE <int>, ConstNAME <chr>, Year <dbl>,
#   Allocation <dbl>

```

Step 4: Summarize Allocations Computes county-level totals per year.

```

County_Summary <- NCDF_long %>%
  group_by(County, Year) %>%
  summarise(Total_Allocation = sum(Allocation, na.rm = TRUE), .groups = "drop")
head(County_Summary);

```

```

# A tibble: 6 x 3
  County   Year Total_Allocation
  <chr>   <dbl>         <dbl>
1 Kajiado 2014      370025209
2 Kajiado 2015      530830055
3 Kajiado 2016      562468559
4 Kajiado 2017      409482759.
5 Kajiado 2018      490948276.
6 Kajiado 2019      545204378.

```

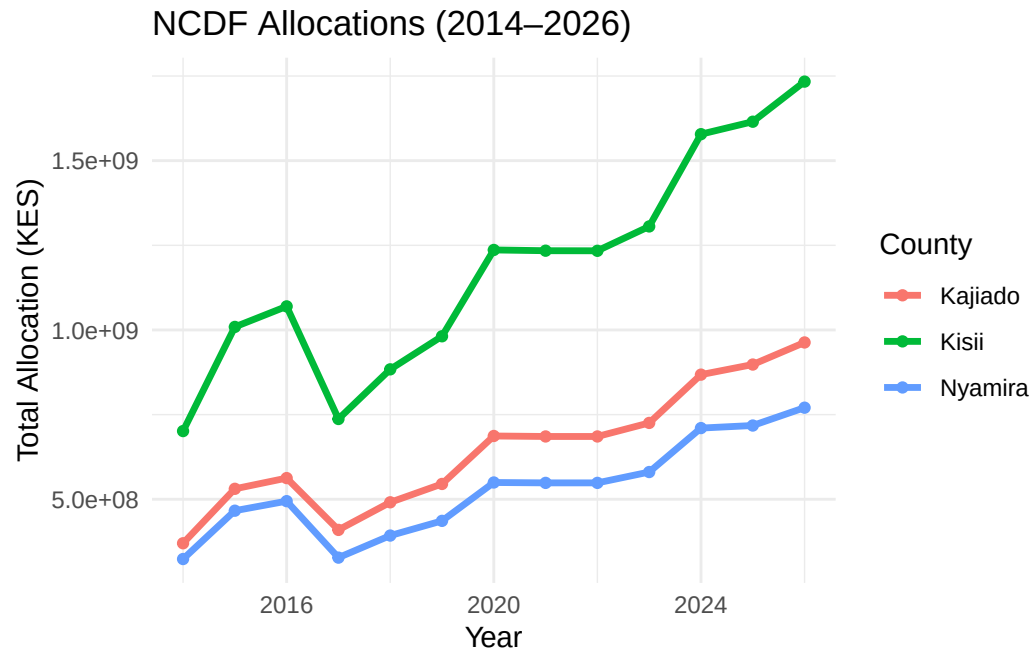
Step 5: Visualization Generates plots to show trends and comparisons.

```

ggplot(County_Summary, aes(x = Year, y = Total_Allocation, color = County)) +
  geom_line(size = 1.2) +
  geom_point() +
  labs(title = "NCDF Allocations (2014-2026)",
       x = "Year", y = "Total Allocation (KES)") +
  theme_minimal()

```

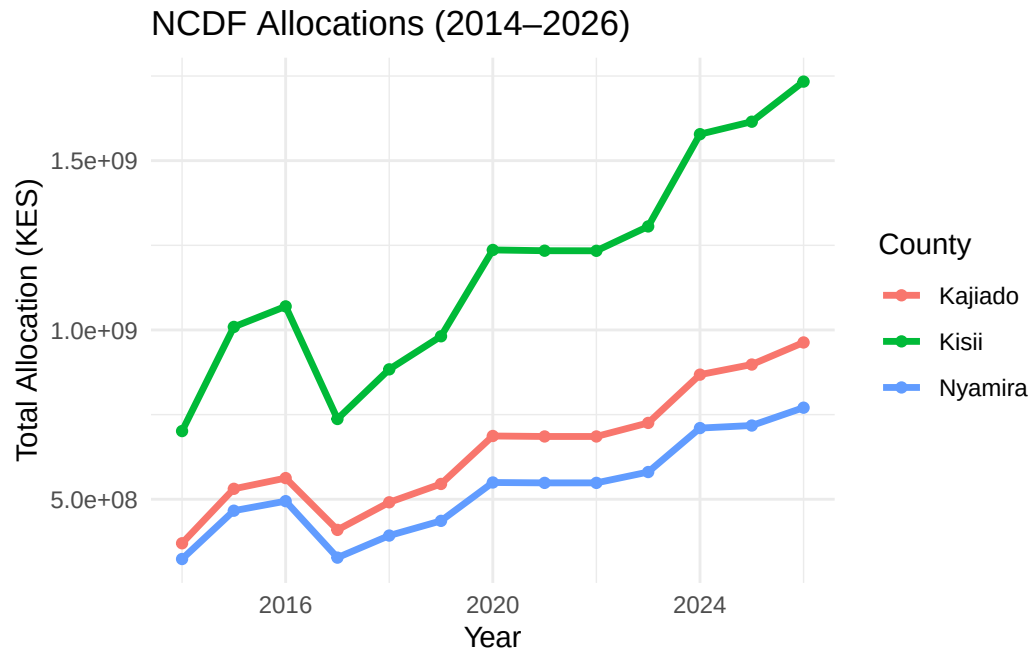
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



Results and Analysis

County-Level Trends

```
ggplot(County_Summary, aes(x = Year, y = Total_Allocation, color = County)) +  
  geom_line(size = 1.2) +  
  geom_point() +  
  labs(title = "NCDF Allocations (2014-2026)",  
        x = "Year", y = "Total Allocation (KES)") +  
  theme_minimal()
```

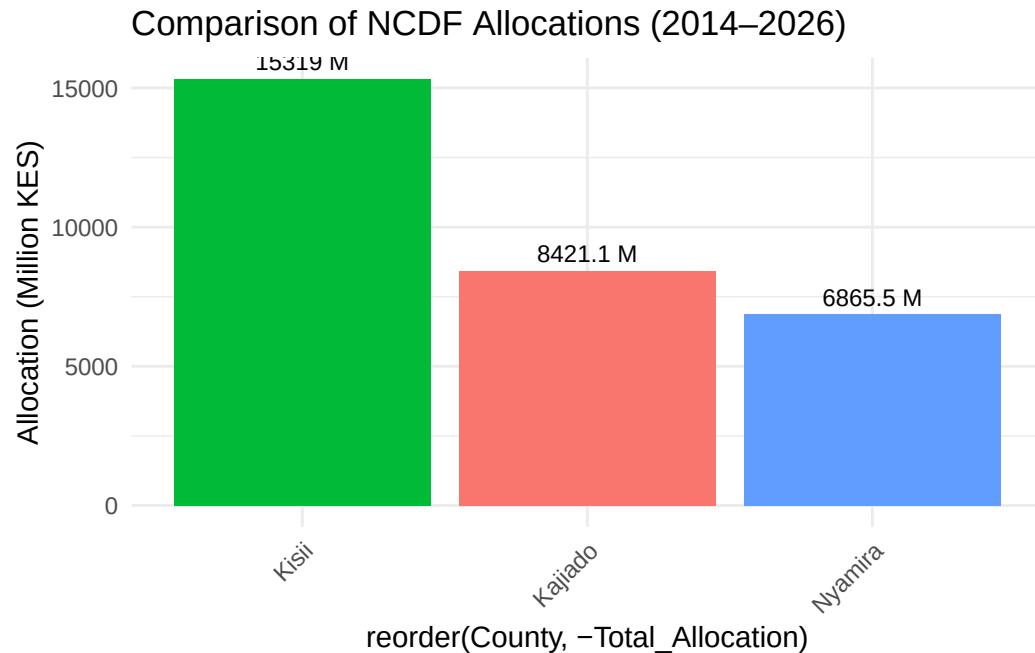


This line plot shows how allocations have evolved over time. Kisii and Nyamira display steady upward trends, while Kajiado's allocations fluctuate more, suggesting variability in funding priorities.

County Comparison (Total Allocations)

```
df <- County_Summary %>%
  filter(Year >= 2014, Year <= 2026) %>%
  group_by(County) %>%
  summarise(Total_Allocation = sum(Total_Allocation, na.rm = TRUE))

ggplot(df, aes(x = reorder(County, -Total_Allocation),
  y = Total_Allocation/1e6,
  fill = County)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = paste0(round(Total_Allocation/1e6,1), " M")),
    vjust = -0.5, size = 3) +
  labs(title = "Comparison of NCDF Allocations (2014-2026)",
    y = "Allocation (Million KES)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none")
```

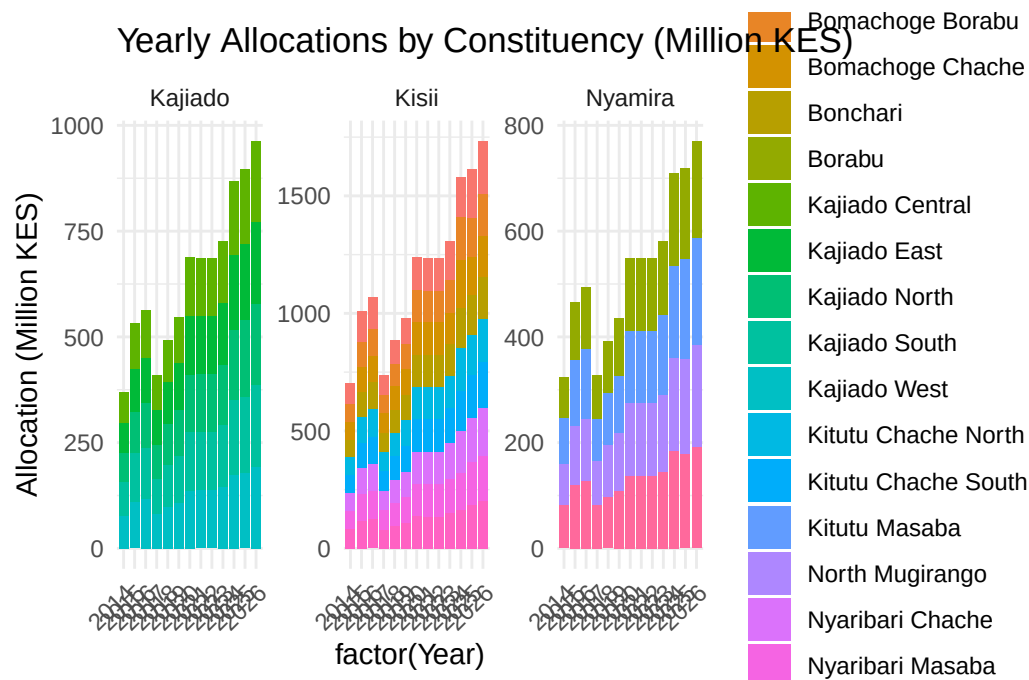



This bar chart compares total allocations across the three counties. It highlights which county received the highest cumulative funding over the study period.

Cross-County Yearly Allocations

```
df <- NCDF_long %>%
  filter(Year >= 2014, Year <= 2026,
         County %in% c("Kisii", "Nyamira", "Kajiado"))

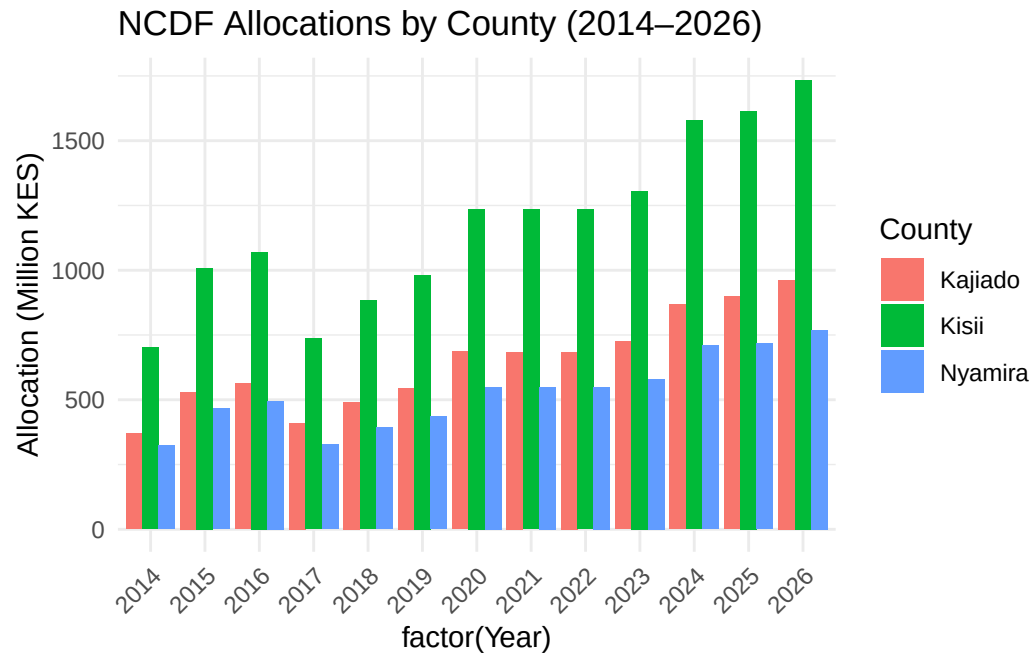
ggplot(df, aes(x = factor(Year),
                y = Allocation/1e6,
                fill = Constituency)) +
  geom_bar(stat = "identity") +
  facet_wrap(~ County, scales = "free_y") +
  labs(title = "Yearly Allocations by Constituency (Million KES)",
       y = "Allocation (Million KES)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



This faceted chart shows yearly allocations broken down by constituency within each county. It reveals which constituencies dominate funding in a given year and highlights intra-county disparities.

Side-by-Side County Comparison

```
ggplot(County_Summary, aes(x = factor(Year),
                             y = Total_Allocation/1e6,
                             fill = County)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "NCDF Allocations by County (2014-2026)",
        y = "Allocation (Million KES)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

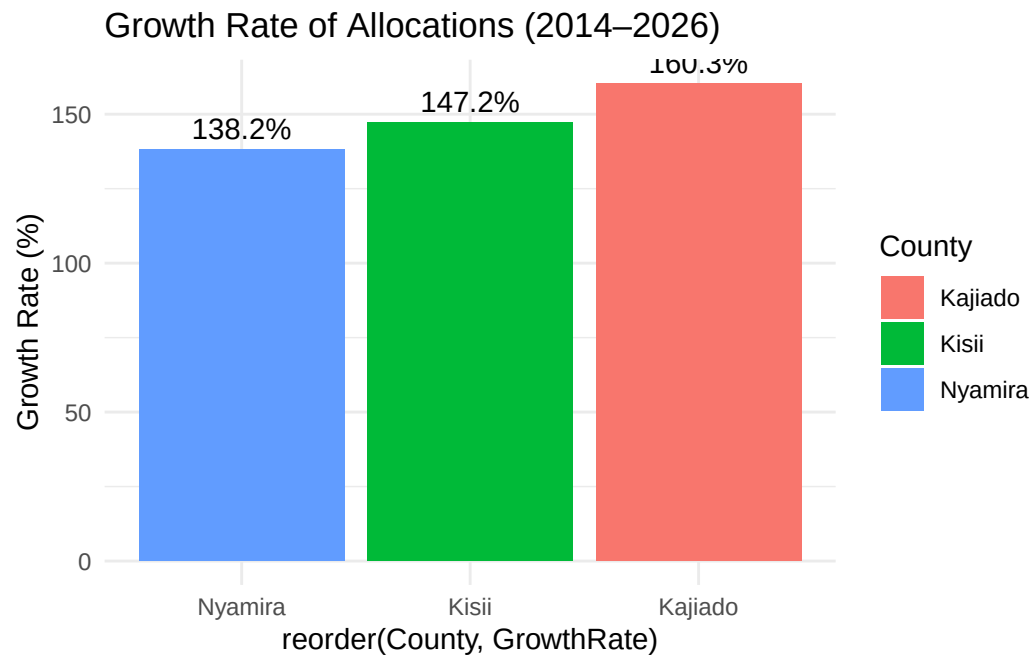


This grouped bar chart compares counties year by year. It makes clear whether one county consistently outpaces the others or whether funding shifts across years.

Growth Rate Comparison

```
growth_df <- County_Summary %>%
  group_by(County) %>%
  summarise(GrowthRate = (last(Total_Allocation) - first(Total_Allocation)) /
              first(Total_Allocation) * 100)

ggplot(growth_df, aes(x = reorder(County, GrowthRate),
                        y = GrowthRate,
                        fill = County)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = paste0(round(GrowthRate,1), "%")),
            vjust = -0.5) +
  labs(title = "Growth Rate of Allocations (2014-2026)",
       y = "Growth Rate (%)") +
  theme_minimal()
```



This chart shows the percentage growth in allocations from 2014 to 2026. It highlights which county experienced the fastest growth in funding, offering insight into changing budget priorities.

Top Constituencies per Year

```
library(knitr)

top_constituencies <- NCDF_long %>%
  filter(County %in% c("Kisii", "Nyamira", "Kajiado"),
         Year >= 2014, Year <= 2026) %>%
  group_by(County, Year) %>%
  slice_max(order_by = Allocation, n = 1, with_ties = FALSE) %>%
  select(Year, County, Constituency, Allocation)

top_constituencies %>%
  kable(caption = "Top Constituency Allocations per County per Year (2014-2026)")
```

Table 1: Top Constituency Allocations per County per Year (2014–2026)

Year	County	Constituency	Allocation
2014	Kajiado	Kajiado South	78964640
2015	Kajiado	Kajiado South	113679933
2016	Kajiado	Kajiado South	120518817
2017	Kajiado	Kajiado North	81896552
2018	Kajiado	Kajiado North	98189655
2019	Kajiado	Kajiado North	109040876
2020	Kajiado	Kajiado North	137367724
2021	Kajiado	Kajiado North	137088879
2022	Kajiado	Kajiado North	137088879
2023	Kajiado	Kajiado North	145087603
2024	Kajiado	Kajiado Central	175361810
2025	Kajiado	Kajiado North	179898372
2026	Kajiado	Kajiado North	192613248
2014	Kisii	Bobasi	88552495
2015	Kisii	Bobasi	128205784
2016	Kisii	Bobasi	136032885
2017	Kisii	Bonchari	81896552
2018	Kisii	Bonchari	98189655
2019	Kisii	Bonchari	109040876
2020	Kisii	Bonchari	137367724

Year	County	Constituency	Allocation
2021	Kisii	South Mugirango	137223079
2022	Kisii	Bonchari	137088879
2023	Kisii	Bobasi	165705316
2024	Kisii	Bomachoge Chache	201666082
2025	Kisii	Bobasi	206358248
2026	Kisii	Bobasi	221505235
2014	Nyamira	Kitutu Masaba	87769627
2015	Nyamira	Kitutu Masaba	127019718
2016	Nyamira	Kitutu Masaba	134766129
2017	Nyamira	Kitutu Masaba	81896552
2018	Nyamira	Kitutu Masaba	98189655
2019	Nyamira	Kitutu Masaba	109040876
2020	Nyamira	Kitutu Masaba	137367724
2021	Nyamira	Kitutu Masaba	137088879
2022	Nyamira	Kitutu Masaba	137088879
2023	Nyamira	Kitutu Masaba	151960174
2024	Nyamira	West Mugirango	184129901
2025	Nyamira	Kitutu Masaba	188414052
2026	Nyamira	Kitutu Masaba	202243910

This table identifies the top-funded constituency in each county for every year. It shows which

constituencies consistently attract the highest allocations, and whether dominance shifts over time. Comparing across counties also reveals if one constituency stands out nationally among the three.

Findings

The analysis of NCDF allocations across Kisii, Nyamira, and Kajiado between 2014 and 2026 reveals several key insights:

County-level trends - Kisii and Nyamira show steady increases in allocations, while Kajiado exhibits more variability, reflecting shifting priorities or budget constraints.

Constituency comparisons - Certain constituencies consistently receive higher allocations, suggesting prioritization or larger project scopes.

Growth rates - Growth rates vary widely, with some constituencies showing strong upward trends while others remain stagnant.

Top allocations - Each year, one constituency emerges as the top recipient in each county. Comparing across counties shows that Kisii constituencies often dominate, though Nyamira and Kajiado occasionally surpass them in specific years.

Identified Gaps

Despite the dashboard's strength in tracking financial allocations, several limitations remain:

Project-level data is missing — the dashboard does not show what projects were funded (schools, health centers, water points, roads).

Completion status is not captured — stakeholders cannot assess whether projects were completed, ongoing, or stalled.

Impact assessment is absent — the quality, sustainability, and community impact of funded projects are not evaluated. Accountability metrics such as planned vs. actual outputs, unit costs, and delays are not integrated.

Recommendations

To strengthen accountability and transparency, the following enhancements are proposed:

Integrate project-level data - Extend the dataset to include project descriptions, categories, start/end dates, and completion status. Link allocations to specific outputs (e.g., “KES 20M → 3 classrooms built in Kisii South”).

Web scraping for project information - Use tools like rvest in R to collect project lists from NCDF Board, county portals, and parliamentary reports.

Dashboard enhancements - Add a Projects tab showing allocations linked to outputs. Use interactive maps to show project locations. Include completion status indicators (e.g., green = completed, red = stalled). Accountability metrics - Compare planned vs. actual projects. Track unit costs (e.g., cost per classroom). Highlight delays or discrepancies between allocations and outcomes.

Stakeholder engagement - Provide downloadable reports for community monitoring. Enable citizen feedback loops to validate project completion.

Conclusion

The NCDF dashboard for Kisii, Nyamira, and Kajiado counties provides a strong foundation for financial transparency. The findings show clear allocation trends, county comparisons, and constituency-level disparities. However, allocations alone do not capture the development outcomes.

To achieve accountability, the dashboard must evolve to include project-level data. By integrating project descriptions, completion status, and impact metrics, stakeholders will be able to compare financial inputs with tangible outputs. This transformation will ensure that NCDF allocations are not only tracked as numbers but also evaluated in terms of their real impact on communities like empowering citizens, policymakers, and researchers to monitor progress and demand accountability.